

DRAFT STATEMENT OF WORK

WEAPONS EFFECTS

1.0 BACKGROUND

The Naval Surface Warfare Center, Carderock Division (NSWCCD), Survivability and Weapons Effects Division (Code 66) is responsible for theoretical and applied research efforts to extend the understanding of the response of US Navy platforms to threat weapon engagements. These engagements include underwater explosions (UNDEX), air explosions (AIREX), and penetrating weapons with modeling & simulation (M&S) of UNDEX and AIREX events being a major part of this mission that requires unique software applications and expertise.

2.0 OBJECTIVE

The contractor shall provide scientific and technical services with experts in UNDEX and AIREX analysis of undersea and surface vessels to develop and utilize specialized software codes that: 1) Aid in the selection of live fire events and provide support to live fire test and evaluation programs; 2) Simulate an underwater shock environment from a far field or near miss location; 3) Simulate the response of Navy platforms to UNDEX and AIREX for far field, near miss, contact, and internally detonating weapon attacks; 4) Check the integrity of a computational models; 5) Solve finite element model's elastic-plastic response and reduced order engineering tools to UNDEX and AIREX events; and 6) Simulate a nuclear underwater shock environment or nuclear air blast environment from a far field location. The technical areas in this Task Order (TO) shall include comprehensive software engineering, development, and enhancement, finite element and reduced order model construction to include deactivation diagrams and quality assurance, numerical model and algorithm review and verification, software application training, design yard qualifications submittal review, data analysis, data science, and artificial intelligence / machine learning in order to support the vulnerability design processes for multiple Navy platforms.

3.0 APPLICABLE DOCUMENTS

- MIL-S-901D, MILITARY SPECIFICATION: SHOCK TESTS. H.I. (HIGH-IMPACT) SHIPBOARD MACHINERY, EQUIPMENT, AND SYSTEMS, REQUIREMENTS FOR (17-MAR-1989)

4.0 SCOPE OF WORK

The contractor shall provide specialized scientific, engineering, and technical services in the fields of UNDEX shock qualification, UNDEX damage assessment, AIREX damage assessment, and ship survivability directed towards research, development, test, and evaluation (RDT&E) for various submarine and surface ship programs. The contractor shall provide all services, materials, facilities and equipment necessary to perform specific tasks within the following general tasking areas:

- 4.1 Support the development, implementation, and transition of new technologies involving: testing strategies, transient shock analysis, and analytical methods that improve the ability of ship, components, systems, and personnel to resist the damaging effects of threat weapon attacks. Follow rigorous software development practices, including strict version control, continuous integration, use of code documentation generators to be specified in each Technical Instruction (TI), unit and regression testing, code coverage assessments, and agile management for all tools developed or modified and share code repositories with Government personnel. Support developing, documenting, and execution strategies for verification and validation (V&V) of numerical simulations. Contractor support will range from limited assistance to government engineers on a

task to autonomous execution of tasking with all results reported to the government as required by the TO and associated Contract Data Requirements List (CDRL) deliverables. Guidance and criteria for contractor support will be defined in each individual TI. (CDRL A001, A002, A003)

4.2 Develop Software:

4.2.1 Support in the development of software applications, code, modules, and plugins that will aide in the fields of UNDEX shock qualification, UNDEX damage assessment, AIREX weapons effects evaluation, and ship survivability analysis.

4.3 Develop Training and Documentation Manuals for Software:

4.3.1 Develop detailed software theory manuals, user's manuals, and verification manuals including example problems. (CDRL A003)

4.3.2 Develop and conduct software application training courses for any tools developed in Task Area 4.2 for Government personnel at NSWCCD and/or the contractor's facility. (CDRL A003)

4.3.3 Develop and conduct training related to shock qualification of shock test machines, platforms and vehicles including training on UNDEX theory and methods of analyzing ship and equipment responses to underwater explosion shock. (CDRL A002)

4.4 Software Support:

4.4.1 Provide technical support to users to fix issues that arise post software deployments. Software is to be defined in the individual TI and may include but not be limited to specific weapons effects software such as but not limited to ASAP, NESM, INSiTE, TDC, UERDTools, and SUBVALUE. Maintain logs of technical issues and resolutions and provide quarterly reports tracking resolutions. (CDRL A001, A003, A004)

4.5 Develop Analytical Methods and Various types of Models:

4.5.1 Develop defined finite element and reduced order representations of test articles, platforms, and vehicles including systems and systems deactivation diagrams that appropriately represent and capture their responses to a damaging event due to UNDEX, AIREX, and penetration mechanics, documenting all efforts. (CDRL A001, A002)

4.5.2 Develop or support the creation of or modification to finite element and reduced order models of naval maritime platforms to include structure, components, systems, and system deactivation logic.

4.5.3 Develop defined finite element models and reduced order representations of shock test machines, platforms and vehicles that appropriately represent and capture their responses to a shock event documenting all efforts. (CDRL A002)

4.5.4 Develop and execute strategies for analyses and qualification of external volumes subjected to weapons loads. (CDRL A002)

4.6 Execute Computer Model Simulations and Analyses:

4.6.1 Conduct computational UNDEX analyses of Shock test machines, platforms and vehicles, documenting all efforts. (CDRL A002)

4.6.2 Conduct computational UNDEX, AIREX, and penetration mechanics analyses of test articles, platforms, and vehicles, documenting all efforts. (CDRL A001, A002)

4.6.3 Conduct and document data analyses and research available data and documentation on both physical measurements obtained from Shock Testing and computational data. (CDRL A002)

4.6.4 Support the evaluation of numerical models, reduced order models, and finite element models developed by other contract activities or Government personnel, for accuracy and applicability to shock design and qualification processes, hull integrity evaluations, and live fire test and evaluation assessments including recoverability, documenting all efforts in technical reports. Applicable agreement and use rights to be provided in individual TI defining subject tasking. (CDRL A002)

- 4.6.5 Conduct or support computational weapons effects assessments of threat weapons against naval maritime platforms in all aspects of platform vulnerability and recoverability documenting all efforts. (CDRL Item A002)
- 4.7 Provide support to shock qualification efforts for shipboard equipment in accordance with ship specifications including generation of ship and equipment finite element models, review of technical analyses and reports, and performing finite element and reduced order model analyses. Finite element and reduced order models may be provided by the government or developed by the contractor. Status of finite element and reduced models will be defined in the individual TI(s). (CDRL A002)
- 4.8 Support Testing and Material Procurement:
 - 4.8.1 Provide support to conduct weapons effects testing or other Navy shock qualification testing and measurement efforts including test design, planning, and execution efforts. This includes both physical, on-site support, in addition to support from the contractor's primary place of performance.
 - 4.8.2 Provide support and technical guidance to shock test and trials planning, documenting all efforts. (CDRL A002)
 - 4.8.3 Procure equipment, materials, and information as necessary with consent from Government to provide technical support as defined in the TI(s).

5.0 GOVERNMENT FURNISHED EQUIPMENT/PROPERTY/INFORMATION

The Government shall furnish all necessary Government-Furnished Equipment (GFE) and/or Government-Furnished Property (GFP). Specific GFE/GFP items will be detailed within each corresponding Task Instruction (TI). All requisite information pertaining to the GFE/GFP will be included in the associated TI.

6.0 DATA REQUIREMENTS AND DELIVERABLES

Data to be provided shall be prepared as specified in individual TIs and shall be distributed to the designated Technical Advisor (TA). In addition, a copy of the Monthly Status Report shall be provided to the Contracting Officer's Representative (COR). Below is a table of CRLS for each Task Area.

CDRL	Title	Task Areas
A001	Computer Software Product	4.1 – 4.5
A002	Technical Report – Study Services	4.1 – 4.5
A003	Software User Manual and Support	4.1 – 4.5
A004	Presentation Materials	4.1 – 4.5
A005	Meeting Minutes	4.1 – 4.5
A006	Contracting Officer's Management Report	4.1 – 4.5
A007	Software License Requirements List	4.1 – 4.5

7.0 SECURITY REQUIREMENTS

Security will be in accordance with the terms and conditions cited in the DD Form 254, in Section J, of the TO. The maximum level of security classification required for the work under this contract is SECRET; the classification of work under this contract will be specified in the individual TI. When tasking is at the CUI level, un-cleared US persons may work on the tasking in accordance with relevant security requirements of the specific TI.

- 7.1 Contractor shall be able to send and retrieve classified information and electronic files to/from the government upon request using the Secret Internet Protocol Router Network (SIPRNET), or via FEDEX, or by contractor courier. The maximum duration from the time of request to time of classified information receipt or retrieval shall be no more than 48 hours.
- 7.2 Non-Disclosure
The Contractor shall not disclose any information provided by the Government or developed or relating to the items to be delivered and services to be performed without the PCO's prior approval. All data provided to the Contractor shall be protected from the public. Dissemination or public disclosure includes permitting access to such information by foreign nationals or by any other person or entity, publication of technical or scientific papers, advertising, or any other proposed public release. The Contractor shall physically protect such information to prevent access by any unauthorized person or entity.
- 7.3 Performance in Government Facilities
The Contractor shall comply with the requirements of local instructions regarding performance in Government facilities. All persons engaged in work while on Government property shall be subject to search of their persons (no bodily search) and vehicles at any time by the Government, and shall report any known or suspected security violations to the appropriate Security Department. Assignment, transfer, and reassignment of Contractor personnel shall be at the discretion of the Contractor. However, when the Government directs, the Contractor shall remove from contract performance any person who endangers life, property, or national security through improper conduct. All Contractor personnel engaged in work while on Government property shall be subject to the DoD Standards of Conduct.
- 7.4 Information Security and Computer System Usage
Computer Usage: In accordance with U.S. Navy policy, any personnel, including the Contractor, who utilizes DoD-owned systems, shall assume responsibility for adherence to restrictions regarding internet and e-mail usage. Navy policy prohibits racist, sexist, threatening, pornographic, personal business, subversive or politically partisan communications. All personnel, including the Contractor, are accountable and must act accordingly.
- 7.5 System Monitoring: DoD computer systems are monitored to ensure that the use is authorized, to facilitate protection against unauthorized access, and to verify security procedures, survivability and operational security. During monitoring, information may be examined, recorded, copied, and used for authorized purposes. All information, including personal information, placed on or sent over a DoD system may be monitored. Use of a DoD system constitutes consent to monitoring. Unauthorized use may result in criminal prosecution. Evidence of unauthorized use collected during monitoring may be used as a basis for recommended administrative, criminal or adverse action.
- 7.6 Electronic Spillages
 - 7.6.1 General: Electronic spillages (ES) are unacceptable and pose a risk to national security. An ES is defined as classified data placed on an information system (IS), media or hardcopy document possessing insufficient security controls to protect the data at the required classification level, thus posing a risk to national security (e.g., Sensitive Compartmented Information (SCI) onto collateral, Secret onto Unclassified). The Contractor's performance as it relates to ES will be evaluated by the Government. Electronic spillage reflects on the overall security posture of the Government and a lack of attention to detail with regard to the handling of classified information of IS security discipline. If a Contractor is

determined to be responsible for an ES, all direct and indirect costs incurred by the Government for ES remediation will be charged to the Contractor. The incident will be reflected in the Contractor's performance rating.

- 7.6.2 Spillage Responsibilities: Local command security will continue to be responsible for the corrective action plan in accordance with the security guidance reflected on the DD Form 254. Command security will identify the Contractor facility and contract number associated with all ES during the investigation indicating Contractor support. Command security will notify the contracts division with the Contractor facility name and contract number, incident specifics and associated costs for cleanup. The PCO will be responsible for working with the Contractor to capture the costs incurred during the spillage clean up. The Contractor is also responsible for taking Cybersecurity Awareness training annually via its Facility Security Officer (FSO) as part of the mandatory training requirements. If an ES occurs, additional training will be required to prevent recurrence.

7.7 Portable Electronic Devices (PEDs)

The Contractor shall comply with local command policies and instructions on PEDs.

8.0 TRAVEL

All travel under this effort must be requested of, or authorized by the COR, in writing or by electronic mail, and must show the appropriate order number, the number of people traveling, the number of days for the trip, the reason for the travel, and any high cost or unusual costs expected. The contractor is not authorized to perform any travel that is not in conjunction with this effort. Travel costs shall be in accordance with FAR 31.205-46 and the Joint Travel Regulations. It is estimated that travel to offsite destinations will be required during performance. Locations will be determined by the government and are subject to change. Travel will be specified on individual TIs.

The contractor shall be available to attend meetings at various facilities in support of the TO, which may take place at, but not be limited to, the following locations: Washington D.C.; West Bethesda, MD; Aberdeen, MD; Groton, CT; New York, NY; Newport News, VA; Newport, RI; Philadelphia, PA and various shipyard and test facility locations.

9.0 PERFORMANCE

- 9.1 Place of Performance: Primary place of performance is at the contractor's facilities.
- 9.2 Period of Performance (PoP): The PoP shall be a base year with four, one-year option years.

10.0 SPECIAL QUALIFICATIONS

All services provided by technical software developers and engineers shall be in accordance with all NIST, DoD, DON, NAVSEA, and NSWCCD policies, processes, and procedures, and all applicable commercial standards and regulations.

- 10.1 All contractor personnel supporting this requirement shall be part of the Cyber Workforce (CWF) and the contractor shall provide the Government a list of personnel mapped to their CWF role as part of CDRL A002, and their baseline certificates prior to the start of the period of performance or provide a plan to receive certification within 3 months of the period of performance.
- 10.2 The Contractor shall ensure that all contractor personnel performing under this TO possess or obtain the required operating system/computing environment.
- 10.3 DFARS Clause 252.239-7001 applies to all personnel on this contract. See matrix below for appropriate categories and levels.

Cyber IT - Cyberspace Workforce (CWF) Certification Matrix (aligned with DoDM 8570.01) This matrix identifies contractor training requirements per section eleven of the Statement of Work, in alignment with DFARS clause 252.239.7001, Information Assurance Contractor Training and Certification (JAN 2008) Note: DoD Manual 8140.03 has replaced 8570.01-M as of 15 February 2023.				
Labor Category	SOW Task Area	(Primary Work Role Code) & Title	IAT Level*	Background Investigation
System Developer	Sections 3.1 through 3.6	(621) Software Developer: Develops, creates, maintains, and writes/codes new (or modifies existing) computer applications, software, or specialized utility programs.	IAT II	T5
*NOTE: Acceptable baseline certifications at listed at: https://public.cyber.mil/cw/cwmp/dod-approved-8570-baseline-certifications/				

11.0 TRANSITION PERIOD

- 11.1 A 60-day transition period is scheduled upon award (phase-in) and the 60-day period immediately prior to contract end date (phase-out) to allow for orderly contract transition. To maximize the effectiveness of the transition process, an initial meeting among the Technical Advisors (TAs), Contracting Officer's Representative (COR), PCO (or its delegated representative) and Contract Managers for both contractors shall be held to address phase-in and phase-out requirements and responsibilities not later than five (5) days into the phase in/phase out period. Subsequent meetings will be held as determined by the PCO, however, not less than weekly thereafter with the TA and COR.
- 11.2 **Phase-In**
During the phase-in period, incoming Contractor personnel will be permitted to observe current operations of the outgoing contractor and Government personnel pertaining to the contract. The incoming Contractor shall ensure that its phase-in operations do not interfere with normal operations of the outgoing Contractor, and that maximum effort is made to avoid loss of productivity. By the end of the phase-in period, the incoming Contractor shall have assumed full responsibility for all work requirements under the contract. The incoming Contractor shall be 100 percent staffed by no later than 60 days after contract award.
- 11.3 **Phase-Out**
During the 60-day period immediately prior to the end of this contract, if the incumbent Contractor is not awarded the subsequent contract for these services, the incumbent Contractor shall permit its successor to observe and become familiar with all operations under this contract. The incumbent Contractor shall remain responsible and liable for all requirements of this contract to the final day of this contract. The Contractor shall not defer any requirements for the purpose of avoiding responsibility and transferring such responsibility to the successor Contractor or the Government. The incumbent Contractor shall fully cooperate with the successor contractor and the Government, and shall not interfere with their work or duties to ensure a smooth transition.
- 11.4 **Phase-Out Plan**
The Contractor shall provide the TA, COR, and PCO with a phase-out plan no later than 90 days prior to the contract end date. This plan shall detail disposition action for all GFI and Contractor-held documentation with minimal Government input in the plan, outlining the incumbent Contractor's approach to fulfill the requirements of this contract.

12.0 CONTRACTOR PERSONNEL REQUIREMENTS

12.1 Qualifications and Training. The Contractor shall provide capable personnel with qualifications, experience levels, security clearances, and necessary licenses, certifications, and training as noted in Section J, Attachment X. Training necessary to ensure that personnel performing under this Task Order maintain the knowledge and skills to successfully perform the required functions is the Contractor's responsibility. Training necessary to maintain professional certifications is the Contractor's responsibility and associated costs may not be attributed or directly charged to the Government.

12.2 Identification

Identification of Contractor Employees: In the performance of this contract, Contractor employees shall identify themselves as Contractor personnel by introducing themselves or being introduced as Contractor personnel and by displaying distinguishing badges or other visible identification for meetings with Government personnel or when representatives from another contractor are present. Contractor personnel shall appropriately identify themselves as Contractor employees in telephone conversations and formal and informal written correspondence.

12.3

Contractor Badging: The Contractor shall obtain identification badges from the Government for all Contractor personnel requiring regular access to Government property. The identification badge shall be visible at all times while employees are on Government property. The Contractor shall furnish all requested information required to facilitate issuance of identification badges and shall conform to applicable regulations concerning the use and possession of the badges. The Contractor shall be responsible for ensuring that all identification badges issued to Contractor employees are returned to the appropriate Security Office within 48 hours following completion of the contract, relocation, or termination of an employee, and upon request by the COR or PCO.

13.0 Non-Personal Services/Inherently Governmental Functions

13.3.1. The Government will neither supervise Contractor employees nor control the method by which the Contractor performs the required tasks. The Government will not direct the hiring, dismissal or reassignment of Contractor personnel. Under no circumstances will the Government assign tasks to, or prepare work schedules for, individual Contractor employees. It shall be the responsibility of the Contractor to manage its employees and to guard against any actions that are of the nature of personal services or give the perception that personal services are being provided. If the Contractor feels that any actions constitute or are perceived to constitute personal services, it shall be the Contractor's responsibility to notify the PCO immediately in accordance with FAR clause 52.243-7 Notification of Changes.

13.3.2. Inherently-Governmental functions are not within the scope of this Task Order. Decisions relative to programs supported by the Contractor shall be the sole responsibility of the Government. The Contractor may be required to attend technical meetings for the Government. However, it is not under any circumstances, authorized to represent the Government or give the appearance that it is doing so.